

The 66th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

701.EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Tissue-Specific Myeloablative Conditioning with Total Marrow Irradiation Enhances Gut Microbiome Diversity and Mitigates Gvhd: A Novel Therapeutic Approach

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Background: Traditional systemic myeloablative conditioning methods, such as total body irradiation (TBI) and chemotherapy, severely damage the gastrointestinal (GI) tract and induce graft-versus-host disease (GVHD), negatively impacting patient outcomes in allogeneic transplantation. This gut damage disrupts the microbiota, leading to dysbiosis. Despite numerous efforts, post-treatment interventions have achieved limited success so far. We developed a high-precision total marrow irradiation (TMI) in vivo model, enabling tissue specific dose delivery with higher dose of radiation being delivered to disease sites while sparing other at-risk organs. We previously showed that reducing radiation dose to the gut reduces MadCam1 expression on gut endothelial cells, consequently reducing $\alpha 4\beta 7$ integrin-mediated T cell adhesion to the gut, which may result in less GVHD. Here, we investigated the impact of TMI radiation delivery on gut biodiversity and overall biological responses.

Methods: Balb/c (H2d) mice were subjected to either 8 Gy (4Gy/fraction in 2 fractions in 6 hrs) of TBI or TMI with the latter limiting GI exposure to 4 Gy ($n \geq 4$, duplicate study). Twenty-four hours post-irradiation, the mice received T cell-depleted B6 (H2b) donor bone marrow cells and enriched spleen conventional T cells. We collected fecal samples pre-irradiation, then on days 7 and 14 post-bone marrow transplant (BMT) for microbiome analysis using shotgun metagenomics (Transnetyx) and analyzed on the One Codex platform. We assessed radiation-induced effects on GI using immunofluorescence and electron microscopy and evaluated GI structure-function by analyzing microvilli structure, gut permeability (FITC Dextran), and inflammatory cytokines analysis.

Results: The average microvilli height is significantly ($p=0.003$) reduced in the TBI group (485.93 ± 24 nm) compared to the TMI group (631.69 ± 32.29 nm), suggesting TMI preserves GI structures better than TBI. In addition, FITC-dextran uptake in serum showed increased leakiness in TBI treated mice than TMI mice (Mean Fluorescence; $7.8X 10^5$ vs $6.01X 10^5$, $p=0.0233$) suggesting less damage to GI in TMI treatment. Fecal microbiome analysis showed radiation conditioning alters alpha diversity (species diversity) as indicated by a lower Shannon Diversity Index (SDI median: 1.2 vs ~ 1.0) at post-BMT time-points. However, higher gut microbiome diversity was detected in TMI than TBI-treated mice at Day 7 post-BMT (Median SDI 0.9 vs 1.1). Moreover, TMI-treated mice showed higher relative abundances of the bacterial genera Alistipes and Prevotella, along with their respective species, compared to TBI-treated mice. Fecal samples from TMI-treated mice showed 2-fold decrease ($p=0.034$) in Firmicutes/Bacteroidetes (F/B) ratio and increased SCFA receptors FFAR3/GPR41 (12%, $p=0.03$) compared to TBI, which could promote a better gut barrier integrity and reduce inflammation. TMI-treated mice displayed significantly lower systemic inflammatory cytokine levels (e.g. TNF α , IL-1 α , IL-9, IFN- γ , and IL-17a) as measured by cytokine array, confirmed with Elisa for TNF α (Day 3 and day 7 post BMT, $p=0.002$, $p=0.04$, respectively), compared to TBI-treated mice. Reduction in inflammatory cytokines was correlated with a marked decrease in GVHD.

Conclusions: This study demonstrates that our novel tissue-specific myeloablative conditioning regimen, TMI, effectively preserve gut microbiome diversity and integrity after BMT in the murine model, compared to traditional systemic conditioning methods. TMI's targeted radiation delivery minimizes collateral damage to the gut microbiota by reducing GI exposure, thereby alleviating GVHD through enhanced GI structure-function, increased microbiome diversity, reduced systemic inflam-

mation, and altered immune modulation. These findings open new therapeutic avenues that leverage microbiome preservation to improve transplant outcomes. Future research will explore how TMI maintains the gut microbiome, focusing on the roles of different microbial species and their metabolites, and their influence on immune cell recruitment and activation. This understanding will enhance insights into the interactions between the gut microbiome and immune responses post-irradiation, guiding more effective and personalized therapeutic interventions and ongoing clinical trial planning.

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